

LinearSkills Academy

Class 5 — Conditionals (if / elif / else)

1. Basic if Statement

```
age = 20
if age >= 18:
    print("You are an adult")
```

Note: Python does not use curly braces {} — indentation defines the block.

2. if / elif / else

```
marks = 75

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

elif is short for "else if". As soon as one condition matches, the rest are skipped.

3. Nested if

An if statement inside another if — used when conditions depend on each other.

```
age = 20
has_id = True

if age >= 18:
    if has_id:
        print("Entry allowed")
    else:
        print("ID required")
else:
    print("Too young")
```

4. Python vs C++ Syntax Comparison

Feature	Python	C++
Condition wrap	if age >= 18:	if (age >= 18) {
Block start	Colon : + indentation	Curly brace {

Block end	No symbol (indentation ends block)	Curly brace }
else if	elif	else if
Boolean values	True / False (capital)	true / false (lowercase)
Semicolons	Not needed	Required at end of line ;

Side-by-side example:

Python	C++
<pre> if x > 5: print("big") elif x == 5: print("equal") else: print("small") </pre>	<pre> if (x > 5) { cout << "big"; } else if (x == 5) { cout << "equal"; } else { cout << "small"; } </pre>

5. Common Mistake ■

```

age = 20
if age >= 18
    print("Adult")    # Error! Missing colon ':'

```

Fix: always add a colon : after the condition.